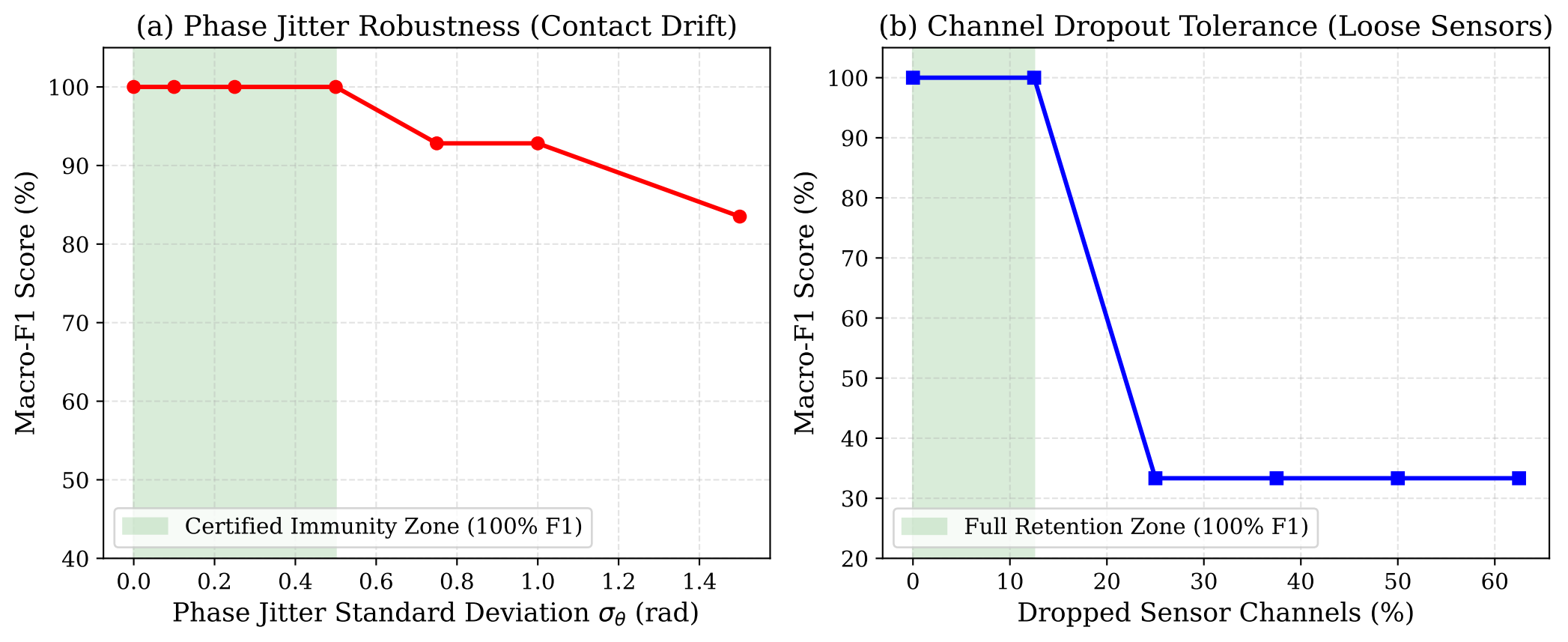
(a) Phase Jitter Robustness (Contact Drift)
Macro-F1 Score (%)
Phase Jitter Standard Deviation σ_θ (rad)
Certified Immunity Zone (100% F1)
100
90
80
70
60
50
40
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
(b) Channel Dropout Tolerance (Loose Sensors)
Macro-F1 Score (%)
Dropped Sensor Channels (%)
Full Retention Zone (100% F1)
100
90
80
70
60
50
40
30
20
0
10
20
30
40
50
60